

2 Description

The STM32C59xxx devices are general purpose microcontrollers family (STM32C5 series) based on the high-performance Arm® Cortex®-M33 32-bit RISC core. They operate at a frequency of up to 144 MHz.

The Cortex®-M33 core features a single-precision floating-point unit (FPU) that supports all the Arm® single-precision data-processing instructions and all the data types.

The Cortex®-M33 core also implements a full set of digital signal processing (DSP) instructions and a memory protection unit (MPU) that enhances the application security.

The devices embed high-speed memories (up to 1 Mbyte flash memory and 128-Kbyte SRAM), one Octo-SPI memory interface, and an extensive range of enhanced I/Os, peripherals connected to three APB buses, three AHB buses, and a 32-bit multi-AHB bus matrix.

The devices feature several protection mechanisms for embedded flash memory and SRAM: readout protection, write protection, and hide protection areas.

The devices embed several peripherals reinforcing security:

- HASH hardware accelerator
- Hardware unique key (HUK)
- True random number generator
- Public key accelerator (PKA) for ECDSA signature verification

The devices offer three 12-bit ADCs, one DAC channel, one comparator, a low-power RTC, four 32-bit general-purpose timers, two 16-bit PWM timers dedicated to motor control, four 16-bit general-purpose timers, two 16-bit basic timers, and one 16-bit low-power timer.

The devices also feature standard and advanced communication interfaces such as:

- Two I²Cs
- One I³C
- Three SPIs with muxed full-duplex I2S
- Four USARTs, three UARTs, and one low-power UART
- Two FDCANs
- One USB full-speed
- One Ethernet MAC interface and one single pair Ethernet

The devices operate in the -40 to +105 °C (+140 °C junction) temperature ranges from a 2.7 to 3.6 V power supply.

A comprehensive set of power-saving modes allows the design of low-power applications.

The devices offer multiple packages from 32 to 144 pins.

See [Table 1](#) for the list of peripherals available for each part number.

Table 1. Device features and peripheral counts

Peripherals	STM32C59xKxT	STM32C59xKxU	STM32C59xCxT STM32C59xCxU	STM32C59xRxT	STM32C59xMxT	STM32C59xVxT	STM32C59xZxT
Number of pins	32	32	48	64	80	100	144
Flash memory (Kbytes)	512/1024						
SRAM (Kbytes)	256						
ICACHE (Kbytes)	8						
Data flash memory (Kbytes)	64						
One-time-programmable (Kbytes)	4.5						
Octo-SPI interface	1						